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Research Paper Report

Research Article:

Mel-MViTv2: Enhanced Speech Emotion Recognition With Mel Spectrogram and Improved Multi scale Vision Transformers

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Problem Definition

Speech emotion recognition:

The task of automatically identifying and classifying emotions from speech signals but mistakenly classified wrong emotions or low efficient

Recognizing emotions from speech signals is a complex task due to the inherent variability in speech patterns and the subjective nature of emotional expression

Limitations of existing methods:

This approach aims to enhance human-computer interaction by analyzing the emotional content of a speaker's speech. It utilizes a method that extracts and classifies prominent features from preprocessed speech signals to determine the speaker's varied emotions. By employing the multilayer perceptron (MLP) classifier, the proposed model is able to classify eight different emotions with an accuracy of 81% on the RAVDESS datasets.

Along with this, visual representations in the form of Mel spectrograms and Mel-STFT spectrograms are used to highlight the frequency and temporal information of the speech signals, providing a more comprehensive representation of the emotional content. The MViTv2 classifier further enhances this approach by introducing multi-scale visual modeling with different stages and pooling attention mechanisms.

Overall, this approach aims to enable more effective and empathetic interactions between humans and computers by accurately analyzing and understanding the speaker's emotions.

Problem Solution

To reduce the mistake

To reduce mistakes in speech emotion recognition, researchers have proposed different approaches that leverage various classifiers and features. One such approach is to use **human-computer interaction (HCI)**. Speech contains two forms of information(textual and emotional). To achieve a harmonious human-computer interaction experience, the computer should automatically recognize the emotional content of speech signals

To reduce the limitations of existing methods :

The proposed method targets the speaker's existence of varied emotions by extracting and classifying the prominent features from a preprocessed speech signal. The multilayer perceptron (MLP) classifier, a widely used supervised learning algorithm, was preferred for classification. The proposed model's performance was compared with that of similar studies,

and the results were evaluated. An overall accuracy of 81% was obtained for classifying eight different emotions by using the proposed model on the RAVDESS datasets.

Mel spectrograms provide visual representations of speech signals, highlighting frequency content relevant for emotion analysis. The Mel-STFT spectrograms capture both the frequency and temporal information of speech signals, providing a more comprehensive representation of the emotional content. The MViTv2 classifier introduces multi-scale visual modeling with different stages and pooling attention mechanisms.

Methodology Used

Literature Search:

The performance evaluation on three diversified speech emotion datasets, namely the Berlin Database of Emotional Speech (Emo-DB), the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), and the Interactive Emotional Dyadic Motion Capture (IEMOCAP) to assess the generalization capabilities of the model.

Study Selection Criteria:

Types of studies: The paper includes experimental studies that evaluate the performance of different methods for speech emotion recognition on various datasets.

Types of participants: The paper includes studies that involve human participants who express different emotions through speech signals.

Types of interventions: The paper includes studies that use different types of features and classifiers for speech emotion recognition, such as handcrafted features, deep learning models, or hybrid models

Types of outcomes: The paper includes studies that report the accuracy, precision, recall, F1-score, or confusion matrix of speech emotion recognition methods on different datasets. The

paper excludes studies that do not report any of these outcomes, or that report irrelevant outcomes for speech emotion recognition.

Quality Assessment:

The methodological quality of the selected studies was accessed using the detecting and analyzing emotional information conveyed through speech signals.

Data Sets:

well-established speech emotion recognition dataset consisting of 535 audio samples from ten professional German speakers, including an equal representation of 5 male and 5 female actors. covers 7 distinct emotions, namely anger, boredom, neutral, happiness, anxiety, sadness, and disgust.

Another widely recognized dataset employed in this evaluation is the RAVDESS [15]. It comprises 1440 audio samples in the English language, recorded by 24 professional actors, with an equal distribution of 12 male and 12 female speakers. RAVDESS covers a wide spectrum of emotions, encompassing neutral, calm, happy, sad, angry, fearful, disgust, and surprised

Algorithms Used:

The Mel Spectrogram with Short-Time Fourier Transform (Mel-STFT) is a powerful audio visualization technique that converts audio signals into visual representations.

The Improved Multi scale Vision Transformers (MViTv2) are an upgrade to MViTv1 that introduces different stages for multi-scale visual modeling, instead of single-scale blocks in Vision Transformer

Tools used:

The primary technology used in this research Speech Recognition technology

SER Software: Applications and software specifically designed for creating Speech Recognitions.

classifiers: the Convolutional Neural Networks (CNNs) , Bi-directional Long Short-Term Memory (Bi-LSTM) classifier and Improved Multi scale Vision Transformers (MViTv2).

spectrograms : short-time Fourier transform (STFT), Mel, Mel-frequency and Mel-STFT spectrogram

Research tools:

Literature Review Tools:

The Berlin Database of Emotional Speech (Emo-DB), the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), and the Interactive Emotional Dyadic Motion Capture (IEMOCAP).

Data Analysis:

The effectiveness of speech emotion recognition using the MViTv2 model was evaluated by comparing four different spectrograms on three datasets: Emo-DB, RAVDESS, and IEMOCAP.

Hyper parameter tuning is a critical process aimed at determining the optimal hyper parameter settings for the Mel-MViTv2 method.

Results obtained

To create an effective SER model, it is necessary to tackle the difficult tasks of creating a suitable algorithm to obtain important speech characteristics that enhance the model performance and overcoming the challenges associated with obtaining and interpreting speech features to identify emotions, which are the primary obstacles to the development of an SER model.

- This Article presents a speech emotion recognition method, known as “Mel-MViTv2” using Mel-STFT spectrograms and MViTv2 as a classifier.
- By dividing the audio signal into short-time frames using a window function, such as the Hann window, and applying the STFT equation, the frequency content of the audio signal at each time point is obtained
- The resulting magnitude spectrum is then transformed into the Mel scale through triangular filter banks, yielding the Mel-STFT representation. This technique effectively captures acoustic features such as pitch changes, intensity variations, and speech rate, which are relevant for speech emotion classification.

- MViTv2 incorporates relative positional embedding that consider the relative distance between input tokens, enhancing the modeling of token interactions
- The proposed Mel-STFT with MViTv2 model achieved promising results, recording the highest accuracy of 91.51%, 81.75%, and 64.03% on the Emo-DB, RAVDESS, and IEMOCAP datasets, respectively.

Future research:

Speech emotion recognition (SER) is a rapidly evolving field that has gained significant attention in recent years. The current research in this area is focused on developing more accurate and efficient algorithms for detecting emotions from speech signals. Some of the areas that could be explored in future research include:

- **Multimodal emotion recognition:** Combining speech signals with other modalities such as facial expressions, body language, child language and physiological signals to improve the accuracy of emotion recognition.
- **Robustness to noise:** Developing algorithms that can accurately detect emotions even in noisy environments.
- **Real-time emotion recognition:** Developing algorithms that can detect emotions in real-time, which would be useful in applications such as virtual assistants and chat bots.
- **Cross-cultural emotion recognition:** Investigating the effectiveness of existing SER algorithms across different cultures and languages.
- **Emotion recognition for clinical applications:** Developing SER algorithms that can be used for diagnosing and treating mental health disorders.
- **lie detection using SER:** Investigating the effectiveness of existing SER algorithms can be used to found Using speech emotion recognition while tells lies